

# Tariq Akilah

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Third-year mechanical engineering student at UC Irvine, working across aerodynamics, avionics, and vehicle hardware. Aerodynamics Engineer on Anteater Electric Racing, UCI's Formula SAE EV team. This portfolio collects seven projects, each with its objective, method, and measured outcome.



## DISCIPLINE

B.S. Mechanical Engineering, UC Irvine

## FOCUS

Fluid dynamics / Structural analysis

## BASED

San Jose & Irvine, California

## CONTACT

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# About the work

I split my time about evenly between analysis and the shop. On the analysis side that means CFD, CAD, and GD&T; on the build side, welding, FDM hardware, and mechanical assembly. Most of the projects here started as a problem someone needed solved and ended with a part, a dataset, or a flight that could be measured against a target.

I transferred to UC Irvine from Mission College with an associate degree in physics and a 3.88 GPA. At UCI I am an Aerodynamics Engineer on Anteater Electric Racing, developing the car's undertray in SolidWorks and validating it in STAR-CCM+. Before that I led a five-person avionics team on a two-stage high-power rocket and ran a 244-case CFD study on a canard-controlled airframe for a faculty-advised research team. On my own time I earned my NAR Level 1 rocketry certification with an airframe I designed and printed, and built a 40 mph motorized trike on a \$500 budget, teaching myself to weld.

I am looking for a summer 2027 internship in aerodynamics, vehicle dynamics, or structures.

## EDUCATION

B.S. Mechanical Engineering, University of California, Irvine — expected 12/2028  
A.S.-T Physics, Mission College — completed 06/2026

## STANDING

Cumulative GPA 3.88 · US Citizen · English, Arabic

## CAD & DESIGN

SolidWorks, Fusion 360, Onshape, AutoCAD, GD&T and tolerance analysis, engineering documentation

## SIMULATION & ANALYSIS

STAR-CCM+ (CFD), SimScale (CFD), OpenRocket

## FABRICATION

FDM 3D printing, MIG welding, soldering, engineering drawings, mechanical assembly

## PROGRAMMING

MATLAB, C / C++, Python

## CERTIFICATION

NAR Level 1 High-Power Rocketry, National Association of Rocketry

## CURRENT TEAM

Anteater Electric Racing, UCI Formula SAE EV — Aerodynamics Engineer, 07/2026 to present

## LEADERSHIP

Co-Founder & Vice President, Mission Launch Rocketry — grew the first Mission College project-based engineering club to 60+ members, 14 active on its first project, \$5,300 raised. Founder & President, Mission College Muslim Engineering Society — 35+ members, internships landed for two. Secretary, Mission College Associated Student Government — 70+ official agendas and minutes published.

# Undertray Aerodynamics

Raised undertray downforce by diagnosing turbulent flow entering the inlet from the floor's leading edge and reprofiling the inlet and diffuser geometry in SolidWorks, validated in STAR-CCM+ across braking pitch.

Timeframe Jul 2026 to present      Role Aerodynamics Engineer      Tools SolidWorks, STAR-CCM+

## OBJECTIVE

Hit the downforce target set for this design iteration while staying inside the packaging space left by the chassis, suspension, and powertrain, with every surface clearing the FSAE 2027 aerodynamics regulations under Section T.7.

## APPROACH

Traced the low downforce to turbulent flow separating off the floor's leading edge before it reached the inlet, then reprofiled the inlet and diffuser in SolidWorks and swept the design in STAR-CCM+ across 0 to 3 degrees of braking pitch. Built a second, independent simulation template alongside the aerodynamics lead to confirm the team's baseline setup.

## MEASURED OUTCOME

- Undertray downforce raised from **under 50 N to 160 N** in STAR-CCM+, swept across 0–3° of braking pitch.
- Independent simulation template agreed with the team baseline **within 10%**, validating the setup.
- Now integrating into a full-car aero package with the front wing, rear wing, and sidepod subteams across a 35+ member aerodynamics team; **targeting 200–250 N** for the final iteration.

Figures withheld. This is an active FSAE competition program; the team has asked that CFD imagery and design detail stay unpublished while the car is competing.

# Canard Roll-Control CFD

Ran 244 SimScale CFD cases on Thrust Stack, a canard-controlled airframe built for closed-loop roll stabilization, and delivered the roll torque dataset the flight controller runs against.

Timeframe Jun 2026 to Aug 2026    Role Fluid Dynamics Research Intern    Tools SimScale, SolidWorks

## OBJECTIVE

The flight controller had no data relating canard deflection to the roll torque it produces at a given airspeed, so it could not turn a commanded correction into an actual canard angle. The dataset had to cover the full envelope: deflection 0.55 to 9.9 degrees, velocity 33 to 113 m/s.

## APPROACH

Ran a SimScale sweep across the full range. The first canard geometry gave too much torque per degree for fine corrections near small angles, so I scaled the canards about 30% smaller to trade peak torque for resolution. Also designed and manufactured a custom servo housing in SolidWorks, sized to sit flush with the airframe so it added no parasitic drag.

## MEASURED OUTCOME

- **244 CFD cases, ~6,200 core-hours** of simulation across the full envelope.
- **162-point roll torque coefficient dataset** delivered as the lookup table the controller reads against commanded deflection and measured speed; adopted directly into the flight control logic.
- Conversion to angular acceleration held cleanly to a **15-degree** maximum deflection, which now sets the controller's usable range.
- Smaller canard geometry selected for finer control. Rocket scheduled to fly **January 2027** at the Friends of Amateur Rocketry site, Mojave Desert.

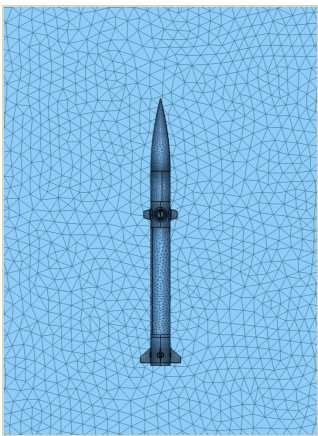


Fig. 01 – SimScale mesh, full airframe

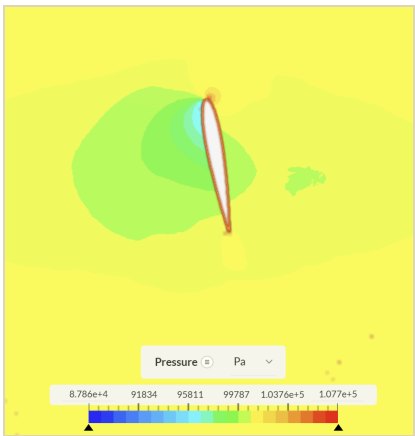


Fig. 02 – Pressure field, canard section

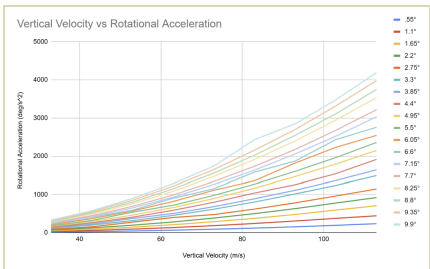


Fig. 03 – Roll acceleration vs. velocity, by deflection

# Two-Stage Rocket Avionics Bay

Led a five-person avionics team through design, integration, and flight test of a two-stage high-power rocket, cutting bay mass while keeping the stability margin and recovery reliable.

Timeframe Jul 2025 to Jun 2026    Role Avionics Lead    Tools Onshape, SolidWorks, OpenRocket, FDM

## OBJECTIVE

Keep the avionics bay light enough that a second stage's added mass did not eat the stability margin, and make the dual-redundant flight computers reliable enough that staging and dual-deploy recovery worked.

## APPROACH

Built the first bay around fitment and hard-landing survival, then reworked the layout to shrink its footprint, swapped in lighter batteries and electronics, and tuned FDM print parameters until mass came down without losing structural integrity, modeling the airframe in OpenRocket alongside that work. The first two launches each cut short; because the bay was built around dual-redundant recovery, the airframe came back intact both times with data to diagnose.

## MEASURED OUTCOME

- Avionics bay mass **reduced 34%**, worth roughly **318 ft** of simulated apogee.
- OpenRocket model held a **1.12-cal stability margin**, predicting apogee **within 7%** of previously measured flight data.
- Dual-redundant flight computers recovered data from **two failed launches**; root causes traced to a faulty barometric reading (fixed with pressure-relief holes) and wind-induced tilt (angle tolerance widened to **45°**).
- Airframe built and cleared for a **final flight simulated at 5,400 ft**.

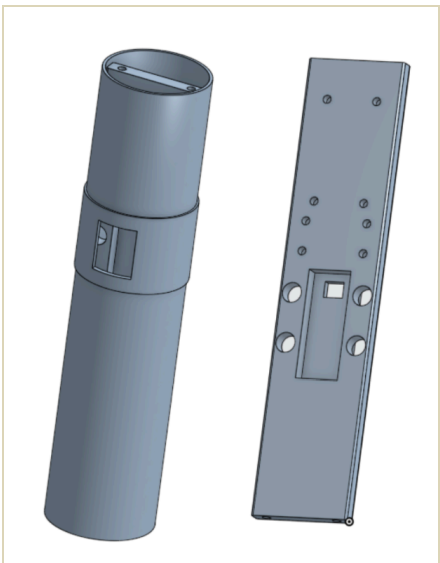


Fig. 01 – Avionics bay, Onshape model

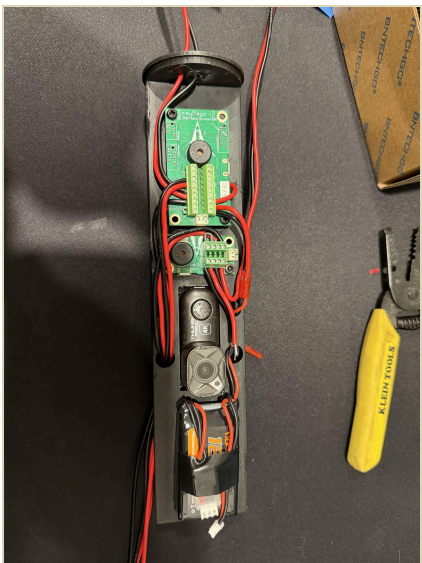


Fig. 02 – Avionics bay sled, as built and wired

# High-Power Rocketry Certification

Designed, FDM-printed, and flew a PETG-CF rocket airframe to earn NAR Level 1 High-Power Certification, holding it under 600 grams without giving up the rigidity a high-power motor demands.

Role Designer and fabricator Tools Fusion 360, OpenRocket, FDM (PETG-CF, Bambu Lab P1S)

## OBJECTIVE

Design, manufacture, launch, and recover a rocket to earn the NAR Level 1 certification, clearing two separate bars: all NAR high-power requirements plus a preflight inspection by a NAR Level 3 member, and a recovery clean enough to leave no flight-critical damage. Self-set target: keep the airframe under 600 grams.

## APPROACH

Modeled the airframe in OpenRocket, then designed the components in Fusion 360 for FDM printing, testing fin geometries, thicknesses, and materials before settling on PETG-CF. The first launch failed on recovery when the parachute detached. Redesigned the fin can from scratch, integrating recovery and motor retention into one part and running the shock cord the full length of the fin can, looped back on itself.

## MEASURED OUTCOME

- Airframe held geometric symmetry under OpenRocket-simulated flight loads at **under 600 grams**.
- Second flight: **stable Mach 0.747 profile to 4,600 ft**, recovered **without critical damage**.
- Flight plus a NAR Level 3 preflight inspection (structural rigidity, construction and assembly, motor retention and tolerancing) earned **NAR Level 1 High-Power Certification**.

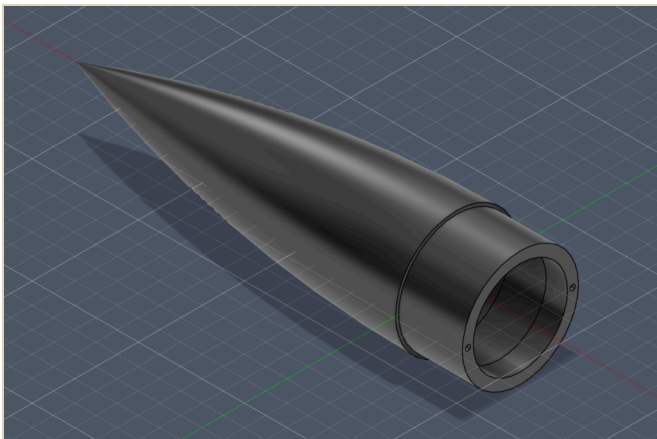


Fig. 01 – Nose cone, Fusion 360



Fig. 02 – Airframe and recovery gear, post-flight



# Motorized Trike — Full Vehicle Build

Built a running motorized trike from recycled material on a \$500 budget and a five-week deadline, teaching myself MIG welding and rebuilding a seized engine to do it.

Timeframe 2024, five-week build    Role Designer and fabricator    Tools Fusion 360, MIG welding

## OBJECTIVE

Design and build a functioning trike inside a five-week program deadline and under \$500, with a chassis that could hold my own body weight (170 to 180 lb) without major flex, running by the end so it could be presented at a design symposium.

## APPROACH

Modeled the chassis in Fusion 360 for a reference to weld against, then built it from 1.75 mm wall steel square tubing, self-taught MIG welding as I went. The powertrain needed more than expected: the donor 212cc engine had seized, so getting it running meant working through a clogged carburetor, a bad spark plug, and the seized flywheel. Mated a go-kart rear axle to a bicycle front fork and tuned the head angle for steering stability at speed.

## MEASURED OUTCOME

- Chassis held a **180 lb rider** through cruising and lateral drifting loads with **minimal chassis flex**.
- Rebuilt 212cc engine and corrected steering geometry reached a **40 mph top speed**.
- Came in roughly **\$100 over** the \$500 target and past the deadline after the originally sourced wheels did not fit the rear axle and had to be replaced.

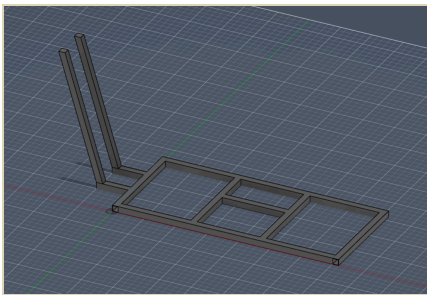


Fig. 01 — Chassis, Fusion 360 model



Fig. 02 — Chassis, as welded



Fig. 03 — Completed trike

# Coilover Suspension Assembly

Reverse-engineered a full coilover unit in Fusion 360 and printed a working PLA assembly, using GD&T-based tolerance analysis across three revisions to get the fit right.

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|      |          |       |                              |        |                                      |
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| Role | Designer | Tools | Fusion 360, PLA FDM printing | Method | GD&T tolerance analysis, 3 revisions |
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## OBJECTIVE

Reverse-engineer a coilover in Fusion 360 to understand how the spring, damper, shock body, adjustment ring, and mounts are shaped and how they define each other's range of motion, then print the assembly in PLA so it was both dimensionally faithful and mechanically functional.

## APPROACH

Worked part by part from product photos and teardown references. Set up a makeshift spring travel in Fusion 360 to confirm the modeled range of motion behaved like a real coilover through compression and rebound and that nothing interfered across that travel. The first printed revision bound on the threads; the next was too loose. Tuned clearances on the threaded and mating surfaces across three revisions.

## MEASURED OUTCOME

- Complete Fusion 360 model of the coilover and a printed PLA assembly built from it.
- After **three tolerance revisions**, the adjustment ring threaded through its full travel smoothly while still holding a secure fit.
- Produced a working feel for how much clearance an FDM fit needs between moving surfaces, carried into later design work.

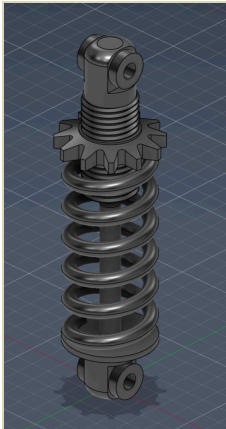


Fig. 01 – Full assembly, Fusion 360



Fig. 02 – Printed PLA components



Fig. 03 – Assembled print



# DAF Tank Redesign — SJSC Wastewater Facility

Designed a one-to-one replacement for a dissolved air flotation tank at the San Jose-Santa Clara Regional Wastewater Facility, working to the real plant's dimensions and delivering the full engineering package.

Role Designer Tools AutoCAD, SolidWorks Review Former facility engineer

## OBJECTIVE

Design a replacement DAF tank that dropped into the existing footprint and connected to the same influent, effluent, air, and skimmer interfaces, so the plant could swap units without reworking surrounding infrastructure. Because it was a direct replacement, the work had to follow the real plant's dimensions and the deliverable had to be a complete package to industry documentation standards.

## APPROACH

Started from the facility's existing plans, pulling tank dimensions and the location of every plant interface. Modeled the tank in SolidWorks and laid the civil and electrical changes into the site plans in AutoCAD. Helped write the addenda, contracts, and change orders to match the format a real facility project would use.

## MEASURED OUTCOME

- Delivered the complete package: SolidWorks model of the replacement tank, updated civil and electrical site plans, and the full set of contract documentation, all built to the existing plant's dimensions and interfaces.
- Design and final package **reviewed by a former engineer of the San Jose-Santa Clara Regional Wastewater Facility.**

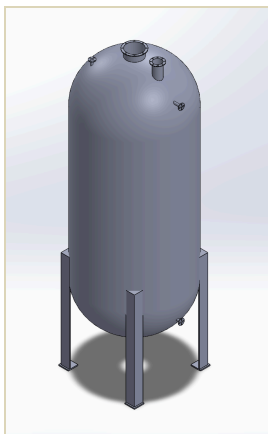


Fig. 01 — Replacement tank model

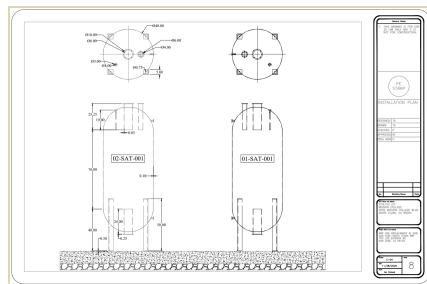


Fig. 02 — Tank drawing, AutoCAD

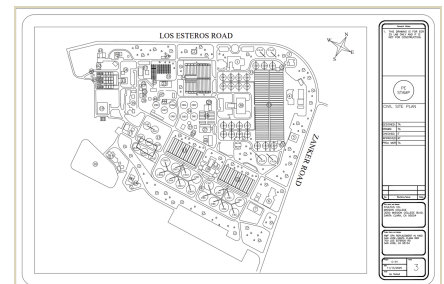


Fig. 03 — Updated civil site plan

TARIQ AKILAH / MECHANICAL ENGINEERING

# Get in touch

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The fastest way to reach me is email, and I answer everything. Happy to talk through any of the work in this portfolio in more detail.

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